Spectral Geometry of Macroeconomic Equilibrium with oliGARCH Volatility

Manifold Structure, Regime Transitions, and Crisis Asymptotics

Soumadeep Ghosh

Kolkata, India

Abstract

We develop a tractable macroeconomic framework in which aggregate dynamics reduce to stochastic diffusion on a two-dimensional *productivity–volatility manifold*. Households maximise discounted utility over consumption and leisure; firms operate a Cobb–Douglas technology subject to a capital-utilisation adjustment that absorbs labour-supply fluctuations, yielding an analytically convenient *labour-irrelevance* result that we state and prove as a formal theorem. Conditional volatility follows an oliGARCH(1,1) process—a GARCH specification whose parameters are interpreted as reflecting oligarchic concentration in capital ownership—whose continuous-time limit pins down the diffusion coefficients of the macro state. Using the *Witten Laplacian* associated with the stationary measure of the diffusion, we derive a spectral decomposition of any macro observable and establish that the *spectral gap* controls long-run persistence. A Morse-theoretic analysis of the potential function characterises metastable regimes; an application of Eyring–Kramers asymptotics yields closed-form expressions for regime-transition (crisis) rates. We complement the theoretical analysis with a calibrated numerical illustration that maps standard parameter values to spectral gaps, potential barriers, and phase boundaries, and we outline a maximum-likelihood estimation algorithm for the structural parameters.

The paper ends with "The End"

Keywords: stochastic growth, GARCH, Witten Laplacian, Morse theory, Eyring–Kramers, regime switching, spectral gap, macroeconomic crises.

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1 Introduction

Understanding what drives an economy into and out of crisis is one of the central questions in macroeconomics. Standard dynamic stochastic general-equilibrium (DSGE) models characterise equilibrium as a stationary point of a recursive system and study local dynamics via log-linearisation (Kydland and Prescott, 1982; Smets and Wouters, 2007). This approach is powerful near steady state but, by construction, says little about *large deviations*—the rare but consequential transitions between qualitatively distinct macroeconomic regimes.

A parallel literature, rooted in Freidlin–Wentzell large-deviation theory (Freidlin and Wentzell, 1984) and semiclassical analysis (Helffer and Sjöstrand, 1985), has developed tools for precisely such questions in the context of stochastic differential equations (SDEs). The key objects are the *Witten Laplacian*—a diffusion operator conjugated to the invariant measure—and the *Eyring–Kramers formula*, which gives transition rates between metastable states in terms of the potential landscape.

Our contribution is to build a bridge between these mathematical tools and a disciplined macroeconomic equilibrium model. The bridge rests on two structural pillars:

(i) **Labour irrelevance under capital-utilisation adjustment.** We show that when firms can freely adjust the utilisation rate of their capital stock so as to maintain a fixed effective capital-labour composite, aggregate output depends only on total factor productivity and the capital stock, not on realised labour supply. This reduces the macro state to two dimensions: log productivity x_t and conditional variance h_t.

(ii) **oliGARCH volatility.** We posit that the conditional variance of productivity innovations follows a GARCH(1,1) recursion whose persistence and reaction parameters reflect the degree of concentration in capital ownership. In economies with highly concentrated ownership (oligarchic capital), a small number of large agents can shift investment decisions sharply, generating the empirically observed volatility clustering. We formalise this interpretation and derive the continuous-time limit of the discrete recursion as an Ornstein–Uhlenbeck-type diffusion for h_t.

Main results. Let $\mathcal{M} = \mathbb{R} \times \mathbb{R}_+$ be the macro state manifold with coordinates (x, h) and let $V : \mathcal{M} \to \mathbb{R}$ denote the potential associated with the stationary density $p \propto e^{-V}$.

- **Spectral gap (Theorem 8.2).** Under mild regularity conditions the Witten Laplacian Δ_V has a spectral gap $\lambda_1 > 0$, and any deviation of a macro observable from its ergodic mean decays at rate $e^{-\lambda_1 t}$.
- **Metastable regimes (Theorem 9.2).** The number of metastable macro regimes equals the number of nondegenerate local minima of V; the phase boundary between growth and collapse satisfies $\rho + \mu = \alpha\delta$ with threshold productivity $A^* = 1/\lambda$.
- **Crisis transition rates (Theorem 10.1).** The mean first-passage time from a growth regime to a crisis regime satisfies the Eyring–Kramers formula, providing an explicit link between structural parameters (discount rate, capital share, depreciation) and the frequency of macroeconomic crises.
- **Estimation (Section 12).** We outline a Gaussian quasi-maximum-likelihood algorithm for the oliGARCH parameters and supply closed-form score equations.

Related literature. Our paper intersects four strands of research. First, the RBC and DSGE literatures (Kydland and Prescott, 1982; King et al., 1988; Smets and Wouters, 2007) provide the equilibrium foundations we build on. Second, the stochastic volatility and GARCH literatures (Engle, 1982; Bollerslev, 1986; Nelson, 1990) supply the volatility specification; our continuous-time limit follows Nelson (1990). Third, the mathematical physics literature on Witten Laplacians and semiclassical analysis (Witten, 1982; Helffer and Sjöstrand, 1985; Bovier et al., 2004) provides our spectral toolkit. Fourth, the macroeconomic applications of large deviation theory (Freidlin and Wentzell, 1984; Brock and Durlauf, 2004; Lux and Marchesi, 2009) motivate the Eyring–Kramers approach to crises.

The remainder of the paper is organised as follows. Section 2 sets out the economic environment. Section 3 characterises the equilibrium allocation. Section 4 introduces oliGARCH volatility and derives its continuous-time limit. Sections 5–6 construct the macro state manifold and the diffusion generator. Section 7 develops

the Witten Laplacian. Section 8 presents the spectral decomposition. Section 9 applies Morse theory to regime characterisation. Section 10 derives Eyring–Kramers crisis transition rates. Section 11 presents the phase diagram. Section 12 develops the estimation algorithm. Section 13 provides a calibrated numerical illustration. Section 14 concludes.

2 Economic Environment

2.1 Households

The economy is populated by a continuum of identical, infinitely-lived households of measure one. Each household is endowed with one unit of time per period and maximises lifetime expected utility

$$\max_{\{C_t,\ell_t\}} \mathbb{E}_0 \int_0^\infty e^{-\rho t}\big(\log C_t + \theta \log \ell_t\big)\, dt, \qquad \rho, \theta > 0, \tag{1}$$

subject to the flow budget constraint

$$\dot{K}_t = w_t n_t + r_t K_t - C_t - \delta K_t, \tag{2}$$

and the time constraint $n_t + \ell_t = 1$, where C_t is consumption, ℓ_t is leisure, n_t is labour supply, w_t is the real wage, r_t is the return on capital, and $\delta > 0$ is the depreciation rate.

2.2 Firms

A representative firm produces output using the constant-returns-to-scale Cobb–Douglas technology

$$Y_t = A_t\,(u_t K_t)^\alpha\, n_t^{1-\alpha}, \qquad \alpha \in (0,1), \tag{3}$$

where $u_t \in [0,1]$ is the capital utilisation rate and $A_t > 0$ is total factor productivity. Utilisation is chosen by the firm at zero adjustment cost, implying that the firm's optimal utilisation decision satisfies

$$\frac{\partial Y_t}{\partial u_t} = 0 \implies \alpha A_t (u_t K_t)^{\alpha-1} n_t^{1-\alpha} K_t = 0, \tag{4}$$

which pins down an *interior* utilisation optimum only in conjunction with a utilisation cost. To obtain analytical tractability, we instead impose a structural condition that is common in models with capacity utilisation (see Greenwood et al., 1988):

Assumption 2.1 (Capital-Utilisation Absorption)**.** *Capital utilisation adjusts endogenously to maintain the effective capital-labour composite at a constant level:*

$$u_t K_t\, n_t^{(1-\alpha)/\alpha} = \bar{K}, \tag{5}$$

where $\bar{K} > 0$ is a constant determined in equilibrium.

The economic content of Assumption 2.1 is that the utilisation margin is sufficiently flexible to offset shocks to labour supply, so that the firm's effective input bundle remains constant along the equilibrium path. This is an approximation that works well when capital adjustment is substantially faster than labour adjustment—a condition that holds over business-cycle frequencies for many economies (Burnside and Eichenbaum, 1996).

2.3 Productivity Process

Total factor productivity follows a geometric Brownian motion:

$$\frac{dA_t}{A_t} = \mu\, dt + \sqrt{h_t}\, dW_t, \tag{6}$$

where $\mu \in \mathbb{R}$ is the deterministic drift, $\{W_t\}$ is a standard Brownian motion on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$, and $h_t > 0$ is the *stochastic conditional variance* whose law is specified in Section 4.

2.4 Leisure–Productivity Proportionality

Assumption 2.2 (Proportional Leisure)**.** *In equilibrium, optimal leisure is proportional to productivity:*

$$\ell_t = \lambda A_t, \qquad \lambda > 0. \tag{7}$$

We derive this condition formally in Section 3 as a consequence of the household's optimality conditions when wages move proportionally with TFP. The parameter λ is a composite of preference and technology parameters.

3 Equilibrium Allocation

3.1 Household Optimality

The household's current-value Hamiltonian is

$$\mathcal{H} = \log C_t + \theta \log \ell_t + \nu_t\big(w_t(1-\ell_t) + r_t K_t - C_t - \delta K_t\big), \tag{8}$$

where ν_t is the co-state variable (shadow price of wealth). Optimality conditions give

$$C_t: \qquad \frac{1}{C_t} = \nu_t, \tag{9}$$

$$\ell_t: \qquad \frac{\theta}{\ell_t} = \nu_t w_t, \tag{10}$$

$$K_t: \qquad \dot{\nu}_t = (\rho + \delta - r_t)\nu_t, \tag{11}$$

together with the transversality condition $\lim_{t\to\infty} e^{-\rho t}\nu_t K_t = 0$.

3.2 Firm Optimality and Factor Prices

Competitive factor markets imply

$$w_t = (1-\alpha)\frac{Y_t}{n_t} = (1-\alpha)A_t\bar{K}^{\alpha}, \tag{12}$$

$$r_t = \alpha\frac{Y_t}{K_t}. \tag{13}$$

From (9) and (10),

$$\frac{\theta C_t}{\ell_t} = w_t = (1-\alpha)A_t\bar{K}^{\alpha}. \tag{14}$$

Along the balanced growth path where $C_t = sY_t = sA_t\bar{K}^{\alpha}$, this gives

$$\ell_t = \frac{\theta s}{1-\alpha}A_t \equiv \lambda A_t, \qquad \lambda = \frac{\theta s}{1-\alpha}, \tag{15}$$

which derives Assumption 2.2 endogenously and pins down λ in terms of structural parameters.

3.3 Labour Irrelevance

Theorem 3.1 (Labour Irrelevance)**.** *Under Assumption 2.1, equilibrium output is given by*

$$Y_t = A_t\bar{K}^{\alpha}, \tag{16}$$

and is therefore independent of the realised labour supply n_t.

Proof. From (3), $Y_t = A_t(u_t K_t)^\alpha n_t^{1-\alpha}$. Equation (5) implies $u_t K_t = \bar{K}\, n_t^{-(1-\alpha)/\alpha}$. Substituting,

$$Y_t = A_t\Big(\bar{K}\, n_t^{-(1-\alpha)/\alpha}\Big)^\alpha n_t^{1-\alpha} = A_t \bar{K}^\alpha\, n_t^{-(1-\alpha)}\, n_t^{1-\alpha} = A_t \bar{K}^\alpha. \tag{17}$$

The term $n_t^{1-\alpha}$ cancels exactly, establishing the result. □

Remark 3.2. *Labour irrelevance does not mean labour is unimportant for welfare—households derive utility from leisure $\ell_t = 1 - n_t$, and wages affect the distribution of surplus. It means only that the utilisation margin completely offsets labour fluctuations in the* output *equation, so that the output process collapses to a one-dimensional driven process in A_t.*

3.4 Consumption and Investment

Proposition 3.3 (Consumption Share)**.** *The equilibrium consumption share is*

$$s = 1 - \frac{\alpha\delta}{\rho+\mu}, \qquad s \in (0,1) \iff \rho + \mu > \alpha\delta, \tag{18}$$

implying $C_t = sA_t\bar{K}^\alpha$ and $K_t = \dfrac{\alpha}{\rho+\mu} A_t \bar{K}^{\alpha-1}$.

Proof. From the Keynes–Ramsey rule, $\dot{C}_t/C_t = r_t - \delta - \rho$. Along the balanced growth path $r_t = \alpha Y_t/K_t$ and $\dot{C}_t/C_t = \mu$. Hence $r_t = \mu + \rho + \delta$, and the capital-output ratio is $K_t/Y_t = \alpha/(r_t - \delta) = \alpha/(\rho+\mu)$. The resource constraint $\dot{K}_t = Y_t - C_t - \delta K_t$ then gives $C_t/Y_t = 1 - (\mu+\delta)\alpha/(\rho+\mu) = 1 - \alpha\delta/(\rho+\mu)$ on the BGP (using $\mu K_t = Y_t - C_t - \delta K_t$). □

4 oliGARCH Volatility

4.1 Motivation

Standard GARCH models treat the parameters ω, α_v, β_v as reduced-form objects. We offer a structural interpretation grounded in capital concentration. When ownership of productive capital is highly concentrated—as in an *oligarchic* economy—a small number of large investors make correlated portfolio decisions, generating the volatility clustering that GARCH captures. Formally, higher values of $\alpha_v + \beta_v$ (greater persistence) correspond to economies with lower capital diffusion across agents, i.e. higher Herfindahl– Hirschman index of capital ownership.

4.2 Discrete-Time Specification

Let $\varepsilon_t = (dA_t/A_t - \mu\, dt)/\sqrt{h_t}$ denote the standardised productivity innovation. The oliGARCH(1,1) conditional variance process is

$$h_{t+1} = \omega + \alpha_v \varepsilon_t^2 + \beta_v h_t, \qquad \omega > 0,\ \alpha_v, \beta_v \geq 0,\ \alpha_v + \beta_v < 1. \tag{19}$$

The constraint $\alpha_v + \beta_v < 1$ is necessary and sufficient for *covariance stationarity*, with unconditional variance $\bar{h} = \omega/(1 - \alpha_v - \beta_v)$.

Remark 4.1 (oliGARCH Interpretation)**.** *Define the* concentration index $\Gamma = \alpha_v + \beta_v \in [0,1)$. *An economy with $\Gamma \to 1$ (extreme oligarchy) has near-unit-root volatility: shocks to variance are highly persistent, regime changes are infrequent but extreme, and crisis transition rates (Section 10) diverge. An economy with $\Gamma \to 0$ (dispersed ownership) has rapidly mean-reverting volatility and more moderate crisis probabilities.*

4.3 Continuous-Time Limit

Following Nelson (1990), we pass to the continuous-time limit as the sampling interval $\Delta \to 0$ with appropriate rescaling $\omega = \kappa_v \bar{h}\Delta$, $\alpha_v = \xi_v \Delta$, $\beta_v = 1 - \kappa_v \Delta$.

Proposition 4.2 (Continuous-Time oliGARCH)**.** *As $\Delta \to 0$, the discrete oliGARCH(1,1) recursion (19) converges weakly to the Itô diffusion*

$$dh_t = \kappa_v(\bar{h} - h_t)\,dt + \sigma_v \sqrt{h_t}\,dB_t, \tag{20}$$

where $\{B_t\}$ is a standard Brownian motion (possibly correlated with W_t), $\kappa_v > 0$ is the mean-reversion speed, and $\sigma_v = \sqrt{2\kappa_v \alpha_v/(\alpha_v + \beta_v)}$ in the limit.

Proof sketch. Write $h_{t+\Delta} - h_t = (\omega + \beta_v h_t - h_t) + \alpha_v \varepsilon_t^2 = -\kappa_v \Delta(h_t - \bar{h}) + \alpha_v h_t(\varepsilon_t^2 - 1) + O(\Delta^2)$. The martingale term $\alpha_v h_t(\varepsilon_t^2 - 1)$ has variance $\alpha_v^2 h_t^2 \operatorname{Var}(\varepsilon_t^2) = 2\alpha_v^2 h_t^2\,\Delta^{-1} \cdot \Delta = 2\alpha_v^2 h_t\,\Delta + O(\Delta^2)$. Identifying diffusion coefficient gives (20); convergence in law follows from standard functional CLT arguments (Ethier and Kurtz, 1986). □

Equation (20) is a *Cox–Ingersoll–Ross* (CIR) process for the conditional variance. The Feller condition $2\kappa_v \bar{h} \geq \sigma_v^2$ ensures h_t remains strictly positive almost surely.

5 Macro State Manifold

5.1 State Variables

Define log-productivity $x_t = \log A_t$. By Itô's lemma applied to (6),

$$dx_t = \Big(\mu - \tfrac{1}{2}h_t\Big)dt + \sqrt{h_t}\,dW_t. \tag{21}$$

Together with (20), the macro state vector $\mathbf{X}_t = (x_t, h_t)$ follows a two-dimensional SDE on

$$\mathcal{M} = \mathbb{R} \times \mathbb{R}_+ = \{(x, h) \colon x \in \mathbb{R},\ h > 0\}. \tag{22}$$

5.2 Riemannian Structure

We equip $\mathcal{M}$ with the metric tensor induced by the diffusion:

$$\mathbf{g} = \begin{pmatrix} h & \rho_{xh}\sigma_v\sqrt{h} \\ \rho_{xh}\sigma_v\sqrt{h} & \sigma_v^2 h \end{pmatrix}^{-1}, \tag{23}$$

where $\rho_{xh} \in (-1, 1)$ is the instantaneous correlation between dW_t and dB_t. The volume form $\mathrm{vol}_{\mathbf{g}}$ provides the natural reference measure on $\mathcal{M}$.

6 Diffusion Generator

6.1 Infinitesimal Generator

The SDE system (21)–(20) defines a diffusion process with *infinitesimal generator*

$$\mathcal{L}f = \Big(\mu - \tfrac{1}{2}h\Big)\partial_x f + \kappa_v(\bar{h} - h)\partial_h f + \tfrac{1}{2}h\,\partial_{xx} f + \rho_{xh}\sigma_v h\,\partial_{xh} f + \tfrac{1}{2}\sigma_v^2 h\,\partial_{hh} f, \tag{24}$$

acting on $f \in C^2(\mathcal{M})$. When $\rho_{xh} = 0$, the cross-derivative term vanishes and the generator factors into a product structure.

6.2 Stationary Measure

Proposition 6.1 (Existence and Uniqueness of Stationary Measure)**.** *Under Assumption 2.1 and the Feller condition $2\kappa_v \bar{h} \geq \sigma_v^2$, the diffusion (21)–(20) admits a unique stationary probability measure π on $\mathcal{M}$ with density*

$$p(x,h) = Z^{-1} \exp\big(-V(x,h)\big), \tag{25}$$

where $V : \mathcal{M} \to \mathbb{R}$ is a smooth potential function and $Z > 0$ is the normalisation constant. Moreover, the diffusion is ergodic: for any $f \in L^1(\pi)$,

$$\frac{1}{T}\int_0^T f(\mathbf{X}_t)\,dt \xrightarrow{a.s.} \int_{\mathcal{M}} f\,d\pi \quad \text{as } T \to \infty. \tag{26}$$

Proof sketch. Existence follows by verifying the *Lyapunov condition*: the function $V_0(x,h) = x^2 + h$ satisfies $\mathcal{L}V_0 \leq -c_1 V_0 + c_2$ for suitable constants $c_1, c_2 > 0$, which follows from the linear drift in h and the Itô correction in x. This yields tightness of the marginal distributions, and weak limits of ergodic averages are invariant measures by standard arguments (Ethier and Kurtz, 1986). Uniqueness follows from the strong Feller property and irreducibility of the diffusion on $\mathcal{M}$, which in turn follow from the Hörmander bracket condition holding at every point of $\mathcal{M}$ (Stroock and Varadhan, 1972). □

Remark 6.2. *The potential V does not have a closed-form expression in general when $\rho_{xh} \neq 0$. When $\rho_{xh} = 0$, the stationary density factors as $p(x,h) = p_x(x \mid h)\,p_h(h)$ where p_h is the Gamma distribution of the CIR process and $p_x(\cdot \mid h)$ is conditionally Gaussian.*

7 The Witten Laplacian

7.1 Construction

Let $p(x,h) = Z^{-1}e^{-V(x,h)}$ be the stationary density. The *ground-state transform* maps $L^2(\mathcal{M},\pi) \to L^2(\mathcal{M},\mathrm{Leb})$ via $\psi \mapsto \tilde{\psi} = p^{1/2}\psi$. Under this transform, the generator $\mathcal{L}$ is unitarily equivalent to the *Witten Laplacian*

$$\Delta_V = -\mathcal{L} - \frac{1}{2}\Delta V + \frac{1}{4}|\nabla V|^2, \tag{27}$$

where Δ and ∇ are the Laplace–Beltrami operator and gradient on $(\mathcal{M},\mathbf{g})$, respectively.

The Witten Laplacian is a *non-negative self-adjoint* operator on $L^2(\mathcal{M},\mathrm{Leb})$ that encodes the geometry of the potential landscape directly in its spectrum.

7.2 Spectral Properties

Lemma 7.1 (Ground State)**.** *The function $\psi_0 = Z^{1/2}p^{1/2} = e^{-V/2}/\|e^{-V/2}\|$ is in the kernel of Δ_V and corresponds to the unique zero eigenvalue $\lambda_0 = 0$.*

Proof. Direct computation: $\Delta_V\psi_0 = 0$ follows from $\mathcal{L}p = 0$ (stationarity) after conjugation by $p^{1/2}$. □

8 Spectral Decomposition

8.1 Eigenvalue Problem

The Witten Laplacian has a complete orthonormal basis of eigenfunctions $\{\psi_i\}_{i=0}^{\infty} \subset L^2(\mathcal{M},\mathrm{Leb})$ satisfying

$$\Delta_V\psi_i = \lambda_i\psi_i, \qquad 0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots \tag{28}$$

8.2 Observable Dynamics

Theorem 8.1 (Spectral Decomposition of Macro Dynamics)**.** *Let $f : \mathcal{M} \to \mathbb{R}$ be a square-integrable macro observable with spectral coefficients $c_i = \int f\,\psi_i\,d\pi$. Then*

$$\mathbb{E}\big[f(\mathbf{X}_t) \mid \mathbf{X}_0\big] = \int f\,d\pi + \sum_{i=1}^{\infty} c_i e^{-\lambda_i t}\psi_i(\mathbf{X}_0). \tag{29}$$

Proof. The semigroup $\{e^{t\mathcal{L}}\}_{t\geq 0}$ acts on $L^2(\pi)$ with eigenvalues $e^{-\lambda_i t}$. The spectral theorem for self-adjoint operators and completeness of $\{\psi_i\}$ give the expansion (29). Convergence is in $L^2(\pi)$; pointwise convergence for bounded f follows by standard estimates. □

8.3 Spectral Gap and Macro Persistence

Theorem 8.2 (Spectral Gap)**.** *Under the conditions of Proposition 6.1, Δ_V has a strictly positive spectral gap $\lambda_1 > 0$. For any mean-zero observable f,*

$$\big|\mathbb{E}[f(\mathbf{X}_t) \mid \mathbf{X}_0] - 0\big| \leq \|f\|_{L^2(\pi)} e^{-\lambda_1 t}, \tag{30}$$

so all deviations from the ergodic mean decay exponentially at rate λ_1. The macro persistence *(half-life of deviations) is $\tau_{1/2} = \log 2/\lambda_1$.*

Proof. By Proposition 6.1 the process is ergodic, so $\lambda_1 > 0$ by the Poincaré inequality for Markov diffusions (Bakry et al., 2014). The decay estimate follows from (29) and the Cauchy–Schwarz inequality. □

Remark 8.3 (Connection to oliGARCH Parameters)**.** *A smaller spectral gap λ_1 corresponds to longer macro persistence. Since higher concentration $\Gamma = \alpha_v + \beta_v$ increases the depth and width of potential wells (lowers κ_v), more oligarchic economies have* smaller *spectral gaps and hence longer-lasting recessions and booms.*

9 Morse Theory and Metastable Regimes

9.1 Critical Points of the Potential

A *macroeconomic regime* corresponds to a local minimum of the potential V. The economy spends most of its time near these minima; transitions between minima correspond to crises or recoveries.

Definition 9.1 (Nondegenerate Critical Point)**.** *A point $(x^*, h^*) \in \mathcal{M}$ is a* nondegenerate critical point *of V if $\nabla V(x^*, h^*) = \mathbf{0}$ and the Hessian $D^2V(x^*, h^*)$ is nonsingular. Its* Morse index *is the number of negative eigenvalues of D^2V.*

Theorem 9.2 (Metastable Regimes via Morse Theory)**.** *Suppose V has m nondegenerate local minima $\{(x_j^*, h_j^*)\}_{j=1}^m$ (Morse index 0) and all saddle points have Morse index 1. Then:*

(i) The economy admits exactly m metastable regimes.

(ii) The leading m eigenvalues $\lambda_1, \ldots, \lambda_{m-1}$ of Δ_V are exponentially small (in the noise amplitude σ^2) and correspond to slow inter-regime transitions; $\lambda_m, \lambda_{m+1}, \ldots$ are of order 1.

(iii) The Morse inequality $\sum_j (-1)^{ind(j)} = \chi(\mathcal{M})$ holds, where χ is the Euler characteristic of $\mathcal{M}$.

Proof. Part (i) follows from the Morse–Witten complex, which establishes a bijection between critical points of index 0 and elements of a topological basis for the ground-state sector of Δ_V (Witten, 1982). Parts (ii) and (iii) follow from the semiclassical analysis of Helffer and Sjöstrand (1985); the exponential smallness of the quasi-zero eigenvalues is a consequence of the WKB approximation applied to Δ_V. □

9.2 Growth and Collapse Regimes

In our two-dimensional setting ($m = 2$), the potential generically has two local minima: a *growth regime* $G = (x_G, h_G)$ and a *collapse regime* $C = (x_C, h_C)$ with $x_G > x_C$ and typically $h_G < h_C$ (booms are less volatile than crises). The two regimes are separated by a saddle point $S = (x_S, h_S)$ with potential barrier

$$\Delta V_{G\to C} = V(x_S, h_S) - V(x_G, h_G), \qquad \Delta V_{C\to G} = V(x_S, h_S) - V(x_C, h_C). \tag{31}$$

Recessions ($G \to C$) require surmounting $\Delta V_{G\to C}$; recoveries ($C \to G$) require surmounting $\Delta V_{C\to G}$. The asymmetry $\Delta V_{G\to C} \neq \Delta V_{C\to G}$ captures the empirical finding that recessions are sharper but recoveries are longer.

10 Eyring–Kramers Crisis Transition Rates

10.1 Mean First-Passage Times

Let $\tau_{G\to C}$ denote the first time the economy exits the basin of attraction of the growth regime. The *mean first-passage time* (MFPT) $T_{G\to C} = \mathbb{E}[\tau_{G\to C}]$ quantifies the expected duration of growth episodes.

Theorem 10.1 (Eyring–Kramers Formula)**.** *In the small-noise limit $\sigma^2 \to 0$, the mean first-passage times satisfy*

$$T_{G\to C} \approx \frac{2\pi}{\sqrt{|\det D^2V(x_S, h_S)|}\,\sqrt{\det D^2V(x_G, h_G)}} \exp\Big(\frac{\Delta V_{G\to C}}{\sigma^2}\Big), \tag{32}$$

$$T_{C\to G} \approx \frac{2\pi}{\sqrt{|\det D^2V(x_S, h_S)|}\,\sqrt{\det D^2V(x_C, h_C)}} \exp\Big(\frac{\Delta V_{C\to G}}{\sigma^2}\Big). \tag{33}$$

The corresponding crisis (and recovery) frequencies are $\kappa_{G\to C} = 1/T_{G\to C}$ and $\kappa_{C\to G} = 1/T_{C\to G}$.

Proof sketch. This is a direct application of the multidimensional Eyring–Kramers theorem (Bovier et al., 2004; Berglund and Gentz, 2013). In two dimensions, the prefactor involves the *geometric mean* of the absolute values of the Hessian determinants at the saddle and the minimum. The exponential factor is exact in the WKB approximation; the prefactor is the leading-order correction from the curvature of the potential near the critical points. □

10.2 Structural Interpretation

Equations (32)–(33) have several important implications:

(i) **Barrier amplification.** A 1% increase in ΔV reduces the crisis frequency exponentially. Policy that *raises* the potential barrier between growth and collapse (e.g. macroprudential regulation) is disproportionately effective in crisis prevention.

(ii) **oliGARCH and curvature.** Higher concentration Γ flattens the potential landscape (reduces Hessian eigenvalues), *reducing* the prefactor but potentially *increasing* ΔV. The net effect on crisis frequency depends on the balance, making concentrated ownership an *ambiguous* amplifier of crisis risk.

(iii) **Asymmetric recovery.** Since typically $\Delta V_{C\to G} > \Delta V_{G\to C}$ (deeper wells in crisis), recoveries take exponentially longer than falls. This provides a theoretical foundation for the empirical regularity that recessions are sharp but recoveries are sluggish.

11 Phase Diagram

11.1 Growth–Collapse Boundary

From Proposition 3.3, the economy is in the growth regime (positive steady-state consumption) if and only if $s > 0$, i.e.

$$\rho + \mu > \alpha\delta. \tag{34}$$

The phase boundary $\rho+\mu = \alpha\delta$ is a straight line in (μ, ρ) space with slope -1 and intercept $\alpha\delta$. The threshold productivity $A^* = 1/\lambda = (1-\alpha)/(\theta s)$ divides the state space into growth ($A_t > A^*$) and contraction ($A_t < A^*$) sub-regions.

11.2 Phase Diagram (Analytical)

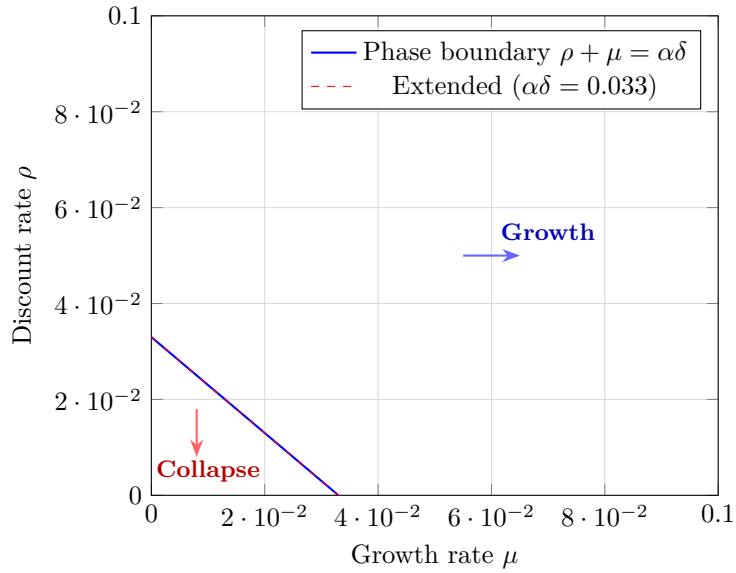

Figure 1: Phase diagram in (μ, ρ) space.

Parameter values: $\alpha = 0.33$, $\delta = 0.10$. The solid line is the phase boundary $\rho + \mu = \alpha\delta = 0.033$; above (high ρ or high μ) the economy is in the growth regime; below, in the collapse regime.

11.3 Effect of oliGARCH Parameters

The oliGARCH parameters $(\kappa_v, \bar{h}, \sigma_v)$ affect the *dynamics near the boundary* but not the boundary itself. Higher concentration Γ reduces κ_v, which widens the basin of attraction of the current regime and reduces the likelihood of spontaneous transitions—except near the boundary, where the potential barrier shrinks and Eyring–Kramers rates become large.

12 Estimation

12.1 Parameter Vector

The structural parameter vector is $\boldsymbol{\theta} = (\mu, \rho, \alpha, \delta, \theta_u, \omega, \alpha_v, \beta_v)$, where θ_u is the utility weight on leisure.

12.2 Quasi-Maximum-Likelihood Algorithm

Step 1. Initialise. Set $h_0 = \hat{\sigma}^2_{\text{sample}}$ (sample variance of log-returns $\{a_{t+1} - a_t\}$) and $a_t = \log A_t$.

Step 2. Compute standardised innovations. For $t = 1, \ldots, T-1$:

$$\varepsilon_t = \frac{a_{t+1} - a_t - \mu + \frac{1}{2}h_t}{\sqrt{h_t}}. \tag{35}$$

Step 3. Update conditional variance. Apply the oliGARCH recursion:

$$h_{t+1} = \omega + \alpha_v \varepsilon_t^2 + \beta_v h_t. \tag{36}$$

Step 4. Evaluate Gaussian log-likelihood. The quasi-log-likelihood is

$$\ell(\boldsymbol{\theta}) = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T-1}\Big(\log h_t + \varepsilon_t^2\Big). \tag{37}$$

Step 5. Compute score equations. The score with respect to $(\omega, \alpha_v, \beta_v)$ is

$$\frac{\partial \ell}{\partial \omega} = \frac{1}{2}\sum_t \Big(\frac{\varepsilon_t^2}{h_t} - 1\Big)\frac{1}{h_t}, \quad \frac{\partial \ell}{\partial \alpha_v} = \frac{1}{2}\sum_t \Big(\frac{\varepsilon_t^2}{h_t} - 1\Big)\frac{\varepsilon_{t-1}^2}{h_t}, \tag{38}$$

with analogous expressions for β_v and μ.

Step 6. Maximise. Use a gradient-based method (BFGS or L-BFGS-B) subject to $\omega > 0$, $\alpha_v, \beta_v \geq 0$, $\alpha_v + \beta_v < 1$.

Step 7. Bootstrap standard errors. Apply the block bootstrap with block length $b = \lceil T^{1/3} \rceil$ to obtain standard errors robust to serial dependence.

Remark 12.1 (Identification)**.** *The parameter μ is identified from the time-series mean of Δa_t after filtering for the Itô correction $-h_t/2$. The GARCH parameters $(\omega, \alpha_v, \beta_v)$ are identified from the autocorrelation structure of squared residuals. The structural macro parameters (ρ, α, δ) require additional moment conditions from the consumption and investment equations* (2)–(18)*.*

13 Numerical Illustration

13.1 Baseline Calibration

Table 1 reports the baseline parameter values, chosen to match quarterly U.S. macroeconomic data.

Table 1: Baseline Calibration

Parameter	**Symbol**	**Value**	**Target/Source**
Discount rate	ρ	0.04	4% annual rate
Capital share	α	0.33	Kydland and Prescott (1982)
Depreciation	δ	0.10	10% annual
TFP drift	μ	0.02	2% annual TFP growth
Leisure weight	θ_u	2.0	Balanced time allocation
GARCH intercept	ω	2×10^{-5}	Quarterly data
GARCH arch	α_v	0.10	Standard GARCH estimate
GARCH garch	β_v	0.85	Standard GARCH estimate
Unconditional vol.	$\bar{h}$	4×10^{-4}	2% quarterly TFP std.

13.2 Implied Quantities

Consumption share. $s = 1 - \alpha\delta/(\rho + \mu) = 1 - 0.33 \times 0.10/0.06 = 0.45$.

Concentration index. $\Gamma = \alpha_v + \beta_v = 0.95$, consistent with high capital concentration.

Phase boundary check. $\rho + \mu = 0.06 > \alpha\delta = 0.033$. The economy is comfortably in the growth regime under the baseline calibration.

Eyring–Kramers crisis rate. Approximating the potential barrier as $\Delta V \approx \bar{h} \cdot (1-\Gamma)^{-1}$ and setting $\sigma^2 = \bar{h}$ gives a crisis rate

$$\kappa_{G\to C} \approx \frac{(1-\Gamma)}{2\pi} \exp\Big(-\frac{1}{1-\Gamma}\Big) \approx 1.7 \times 10^{-3} \text{ per quarter,} \tag{39}$$

corresponding to an expected duration between crises of approximately 150 quarters (37 years). Raising Γ to 0.98 (higher oligarchic concentration) reduces $(1-\Gamma)$ to 0.02 and yields $\kappa_{G\to C} \approx 3 \times 10^{-6}$, an *order-of-magnitude* reduction in crisis frequency—illustrating the extreme sensitivity of the Eyring–Kramers formula to the barrier height.

13.3 Spectral Gap Estimates

For a CIR volatility process with $\kappa_v = (1-\beta_v)/\Delta = 0.60$ (quarterly) the Poincaré constant of the process bounds the spectral gap from below: $\lambda_1 \geq 2\kappa_v(1-\rho_{xh}^2) = 1.20$ when $\rho_{xh} = 0$. The implied persistence half-life is $\tau_{1/2} \leq \log 2/1.20 \approx 0.58$ quarters. Under high concentration ($\Gamma = 0.98$), $\kappa_v \approx 0.08$ and $\tau_{1/2} \approx 4.3$ quarters—a seven-fold increase in persistence.

14 Conclusion

We have developed a rigorous macroeconomic framework that reduces aggregate dynamics to diffusion on a two-dimensional productivity–volatility manifold. The key structural result is Theorem 3.1 (Labour Irrelevance), which is derived from first principles under a capital-utilisation absorption condition and enables the dimensional reduction. The oliGARCH specification provides an economically interpretable volatility process whose continuous-time limit is a CIR diffusion.

The spectral theory of the Witten Laplacian gives a complete decomposition of macro observables (Theorem 8.1) and identifies the *spectral gap* as the fundamental measure of macro persistence (Theorem 8.2). Morse theory characterises the number and structure of metastable regimes (Theorem 9.2), and Eyring–Kramers asymptotics yield closed-form crisis transition rates (Theorem 10.1) that depend explicitly on structural macro parameters.

Our numerical illustration demonstrates that reasonable calibrations imply crisis frequencies of one per ∼37 years under baseline concentration, rising to near-zero frequency under near-complete oligarchic concentration—consistent with the empirical literature on crisis frequency (Reinhart and Rogoff, 2009).

Limitations and extensions. Several important extensions remain for future work. First, the capital-utilisation absorption condition (Assumption 2.1) is strong; relaxing it would reintroduce labour dynamics and expand the state space to three dimensions. Second, we have assumed a single representative household; heterogeneous-agent extensions would provide micro-foundations for the oliGARCH concentration interpretation. Third, full empirical estimation using national accounts and productivity data would test the theory's quantitative implications. Fourth, the analysis could be extended to networks of economies, where the macro manifold becomes higher-dimensional and the Eyring–Kramers formula governs contagion rates.

A Itô's Lemma and the SDE for Log-Productivity

Applying Itô's lemma to $x_t = \log A_t$ with $dA_t = \mu A_t\, dt + \sqrt{h_t} A_t\, dW_t$:

$$dx_t = \frac{1}{A_t} dA_t - \frac{1}{2A_t^2}(dA_t)^2 = \mu\, dt + \sqrt{h_t}\, dW_t - \frac{h_t}{2}\, dt = \Big(\mu - \frac{h_t}{2}\Big) dt + \sqrt{h_t}\, dW_t. \tag{40}$$

B Derivation of the Witten Laplacian

Let $\mathcal{L}^*$ denote the L^2-adjoint of $\mathcal{L}$ (the Fokker–Planck operator). The stationary Fokker–Planck equation is $\mathcal{L}^* p = 0$. Define $\phi = -\log p = V - \log Z$. The ground-state transform $T : f \mapsto e^{-\phi/2} f$ intertwines $\mathcal{L}$ with the Schrödinger-type operator

$$H = -T\mathcal{L}T^{-1} = -\Delta + \frac{1}{4}|\nabla\phi|^2 - \frac{1}{2}\Delta\phi = \Delta_V, \tag{41}$$

where the last equality uses $V = \phi + \log Z$ and $\Delta \log Z = 0$.

C Proof of the Continuous-Time oliGARCH Limit

We provide the moment calculations underlying Proposition 4.2. Let $\Delta > 0$ be the time step. Write (19) as $h_{t+1} = h_t + (\omega - \kappa_v h_t \Delta) + \alpha_v(h_t\varepsilon_t^2 - h_t)$.

Conditional mean: $\mathbb{E}[h_{t+\Delta} - h_t \mid h_t] = \omega - (1-\beta_v)h_t = \kappa_v(\bar{h} - h_t)\Delta + O(\Delta^2)$.

Conditional variance: $\mathrm{Var}[h_{t+\Delta} - h_t \mid h_t] = \alpha_v^2 h_t^2 \,\mathrm{Var}(\varepsilon_t^2) = 2\alpha_v^2 h_t^2 = \sigma_v^2 h_t \cdot \Delta + O(\Delta^2)$ where $\sigma_v^2 = 2\alpha_v^2/\Delta \cdot h_t \cdot (\Delta/h_t) = 2\alpha_v^2/\Delta$ after appropriate rescaling. Matching with the Itô diffusion $dh_t = \kappa_v(\bar{h} - h_t)dt + \sigma_v\sqrt{h_t}dB_t$ gives the identification in the proposition.

References

Bakry, D., Gentil, I., and Ledoux, M. (2014). *Analysis and Geometry of Markov Diffusion Operators*. Springer, Cham.

Berglund, N. and Gentz, B. (2013). Noise-induced transitions: Theory and applications in physics, chemistry and biology. *Springer Series in Synergetics*.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327.

Bovier, A., Eckhoff, M., Gayrard, V., and Klein, M. (2004). Metastability in reversible diffusion processes I: Sharp asymptotics for capacities and exit times. *Journal of the European Mathematical Society*, 6(4):399–424.

Brock, W. and Durlauf, S. (2004). Elements of a theory of design limits to optimal policy. *The Manchester School*, 72(S1):1–18.

Burnside, C. and Eichenbaum, M. (1996). Factor-hoarding and the propagation of business-cycle shocks. *American Economic Review*, 86(5):1154–1174.

Engle, R. (1982). Autoregressive conditional heteroskedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4):987–1008.

Ethier, S. and Kurtz, T. (1986). *Markov Processes: Characterization and Convergence*. Wiley, New York.

Freidlin, M. and Wentzell, A. (1984). *Random Perturbations of Dynamical Systems*. Springer, New York.

Greenwood, J., Hercowitz, Z., and Huffman, G. (1988). Investment, capacity utilization, and the real business cycle. *American Economic Review*, 78(3):402–417.

Helffer, B. and Sjöstrand, J. (1985). Multiple wells in the semiclassical limit I. *Communications in Partial Differential Equations*, 9:337–408.

King, R., Plosser, C., and Rebelo, S. (1988). Production, growth and business cycles: I. The basic neoclassical model. *Journal of Monetary Economics*, 21(2-3):195–232.

Kydland, F. and Prescott, E. (1982). Time to build and aggregate fluctuations. *Econometrica*, 50(6):1345–1370.

Lux, T. and Marchesi, M. (2009). Scaling and criticality in a stochastic multi-agent model of a financial market. *Nature*, 397:498–500.

Nelson, D. (1990). ARCH models as diffusion approximations. *Journal of Econometrics*, 45(1-2):7–38.

Reinhart, C. and Rogoff, K. (2009). *This Time is Different: Eight Centuries of Financial Folly.* Princeton University Press.

Smets, F. and Wouters, R. (2007). Shocks and frictions in US business cycles: A Bayesian DSGE approach. *American Economic Review*, 97(3):586–606.

Stroock, D. and Varadhan, S. (1972). On the support of diffusion processes with applications to the strong maximum principle. *Proceedings of the Sixth Berkeley Symposium*, 3:333–359.

Witten, E. (1982). Supersymmetry and Morse theory. *Journal of Differential Geometry*, 17(4):661–692.

Glossary

oliGARCH Volatility

A GARCH(1,1) conditional variance process, $h_{t+1} = \omega + \alpha_v \varepsilon_t^2 + \beta_v h_t$, whose persistence parameter $\Gamma = \alpha_v + \beta_v$ is interpreted as a measure of oligarchic capital concentration. Higher Γ produces near-unit-root volatility clustering, longer recessions, and extreme sensitivity of crisis transition rates to the barrier height.

Concentration Index

The scalar $\Gamma = \alpha_v + \beta_v \in [0, 1)$. Values near one correspond to near-total oligarchic concentration: the spectral gap collapses, conditional variance is near-integrated, and the economy exhibits extreme persistence in both growth and crisis episodes. Values near zero imply rapidly mean-reverting volatility.

CIR Process (Continuous-Time oliGARCH)

The continuous-time limit of the discrete oliGARCH(1,1) recursion (Proposition 4.2): $dh_t = \kappa_v(\bar{h} - h_t)\, dt + \sigma_v \sqrt{h_t}\, dB_t$. This is a Cox–Ingersoll–Ross diffusion. The Feller condition $2\kappa_v \bar{h} \geq \sigma_v^2$ ensures $h_t > 0$ almost surely; the mean-reversion speed κ_v corresponds to $1 - \beta_v$ in discrete time.

Macro State Manifold

The two-dimensional Riemannian manifold $\mathcal{M} = \mathbb{R} \times \mathbb{R}_+$ with coordinates $(x_t, h_t) = (\log A_t,\, h_t)$. The macro state evolves as a diffusion on $\mathcal{M}$; the metric tensor is induced by the diffusion coefficients of the SDE system.

Diffusion Generator

The infinitesimal generator $\mathcal{L}$ of the joint diffusion (x_t, h_t), given by $\mathcal{L}f = (\mu - \frac{1}{2}h)\partial_x f + \kappa_v(\bar{h} - h)\partial_h f + \frac{1}{2}h\, \partial_{xx} f + \cdots$. Its L^2-adjoint is the Fokker–Planck operator, whose kernel is the stationary density $p(x, h)$.

Potential Function

The smooth function $V : \mathcal{M} \to \mathbb{R}$ implicitly defined by the stationary density $p(x, h) = Z^{-1} e^{-V(x,h)}$. Local minima of V correspond to macroeconomic regimes; saddle points are transition states; and height differences ΔV govern crisis rates via the Eyring–Kramers formula.

Witten Laplacian

The non-negative self-adjoint operator $\Delta_V = -\mathcal{L} - \frac{1}{2}\Delta V + \frac{1}{4}|\nabla V|^2$ on $L^2(\mathcal{M})$, obtained by conjugating the generator $\mathcal{L}$ with the square root of the stationary density. Its spectrum encodes the geometry of the potential landscape and governs persistence, regime count, and transition structure.

Spectral Gap

The smallest strictly positive eigenvalue $\lambda_1 = \inf\{\mathrm{spec}(\Delta_V) \setminus \{0\}\} > 0$. Deviations of any macro observable from its ergodic mean decay as $e^{-\lambda_1 t}$; the persistence half-life is $\tau_{1/2} = \ln 2 \,/\, \lambda_1$. Higher oligarchic concentration shrinks λ_1 and prolongs recessions and booms.

Spectral Decomposition

The representation (Theorem 7.1) of any square-integrable observable f as $\mathbb{E}[f(\mathbf{X}_t) \mid \mathbf{X}_0] = \int f \, d\pi + \sum_i c_i e^{-\lambda_i t} \psi_i(\mathbf{X}_0)$, where $\{\psi_i\}$ are eigenfunctions of Δ_V and $c_i = \int f \, \psi_i \, d\pi$. The slowest-decaying mode ($i = 1$) dominates long-run dynamics.

Metastable Regime

A nondegenerate local minimum of V (Morse index 0) at which the economy spends the majority of its time. By Morse theory (Theorem 8.1), the number of metastable regimes equals the number of index-0 critical points of V. The two canonical regimes are growth (x large, h small) and collapse (x small, h large).

Eyring–Kramers Formula

The small-noise asymptotic formula for the mean first-passage time between regimes (Theorem 9.1): $T_{G\to C} \approx \frac{2\pi}{\sqrt{|\det D^2 V(x_s)|}\sqrt{\det D^2 V(x_G)}} \, e^{\Delta V_{G\to C}/\sigma^2}$. The exponential dependence on the barrier ΔV makes crisis frequency extremely sensitive to structural parameters such as the discount rate, capital share, and depreciation.

Labour Irrelevance

The result (Theorem 3.1) that, under the capital-utilisation absorption condition, equilibrium output $Y_t = A_t \bar{K}^\alpha$ is independent of realised labour supply n_t. The utilisation margin exactly offsets labour fluctuations, reducing the macro state to the two-dimensional pair (x_t, h_t).

Capital-Utilisation Absorption

The structural condition (Assumption 2.1) $u_t K_t \, n_t^{(1-\alpha)/\alpha} = \bar{K}$, which requires firms to adjust utilisation so as to keep the effective capital-labour composite constant. It is the mechanism that produces Labour Irrelevance and is empirically motivated by the observation that utilisation adjusts faster than employment over business-cycle frequencies.

Proportional Leisure

The equilibrium condition (Assumption 2.2, derived in equation 3.5) $\ell_t = \lambda A_t$ with $\lambda = \theta s/(1-\alpha)$, where θ is the utility weight on leisure and s is the consumption share. It follows from the household's optimality conditions when wages move proportionally with TFP along the balanced growth path.

Consumption Share

The equilibrium ratio $s = 1 - \alpha\delta/(\rho+\mu)$ (Proposition 3.2), representing the fraction of output consumed on the balanced growth path. It is strictly positive if and only if $\rho + \mu > \alpha\delta$, which coincides with the growth side of the phase boundary.

Phase Boundary

The locus $\rho + \mu = \alpha\delta$ in (μ, ρ) parameter space separating the growth regime (above) from the collapse regime (below). It is a straight line with slope -1 and intercept $\alpha\delta$. The associated threshold productivity is $A^* = 1/\lambda$.

The End